

Basic of Computer Networks 4

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'I think, in general, it's clear that most bad things come from misunderstanding, and communication is generally the way to resolve misunderstandings — and the Web's a form of communications — so it generally should be good.'

Tim-Bernes Lee - Creator of the WEB

1 Network model

2 The OSI model

- The OSI model
- OSI model: what for?
- The OSI model: show me the layers!
- More details about the layers

3 The TCP/IP Model

- What is it?
- TCP/IP and OSI Model compared
- TCP/IP layers
- Similarities between OSI & TCP/IP
- Differences between OSI & TCP/IP

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What is a network model?

- It is a method for describing and analyzing data communication networks by breaking the entire set of communication process into a number of layers
- Each layer has a specific function (there will be described later)

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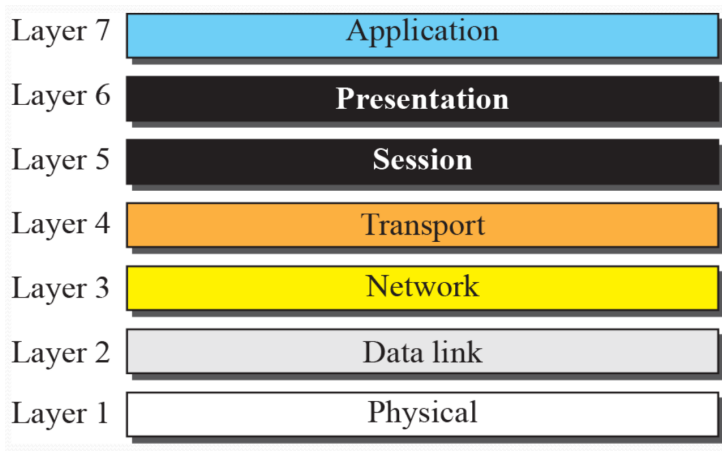
The OSI model was an attempt to unify every network models. Nowadays it is especially a theoretical model people use as a reference model.

- International standard organization (ISO) established a committee in 1977 to develop an architecture for systems communication
- The Open System Interconnection (OSI) reference model is the result of this effort.
- This model allows any two different systems to communicate regardless of their underlying architecture.

The OSI model ...

- ... describes how data flows from one computer, through a network to another computer.
- ... is not a protocol; it is a model for understanding and designing a network architecture that is flexible and robust.
- ... consists of seven separate but related layers, each of which defines a part of the process of moving information across a network.

The OSI model is made of seven layers:

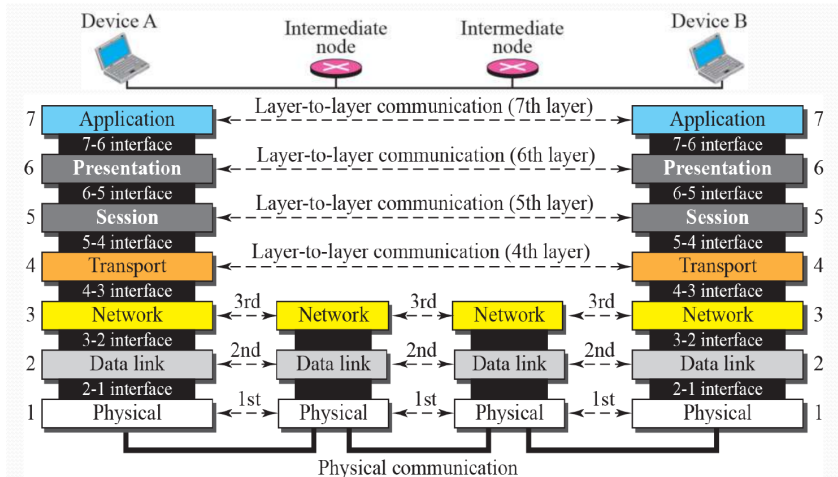


Looks like that, right?



Why so many layers?

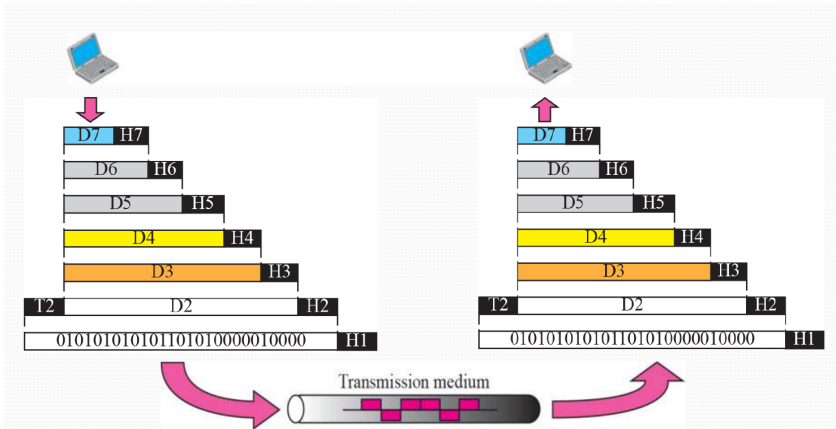
- To reduce the complexity, networks are organized as a stack of layers, one below the other.
- Each layer is responsible for a specific task, it provides a services to an adjacent layer.



Three subgroups of layers:

- Lower layers: physical, data link and network layers are network support layers (responsible for data transmission).
- Middle layer: the transport layer links the two subgroups of layers.
- Higher layers: session, presentation and application layers are user support layers (responsible for application working).

Example of exchange using the OSI model:



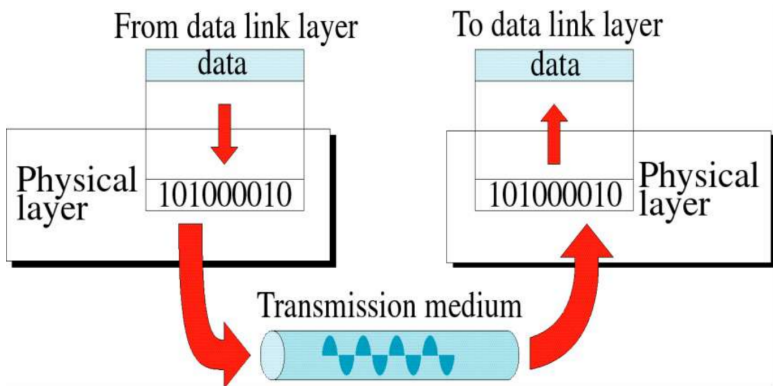
Layer 1: the physical layer

- First layer of the OSI model: all the information of all the other layers go through this layer.
- It is responsible for the actual physical connection between the devices! (ex: cable, WI-FI, etc.)
- Thanks to this layer: individual bits can go from one node (=computer) to the next!

Note

System administrators use to say that most network connection problems occur because of this layer! Typically someone unplugged the cable by mistake!

The physical layer "contains" the data link layer like this:

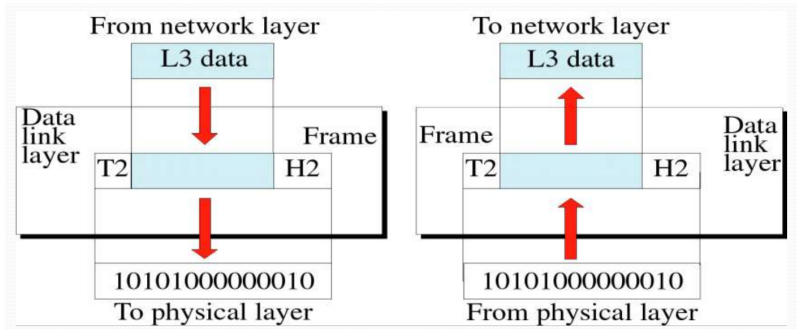


The functions of the physical layer are very low-levels:

- Converting bits to signals
- Bit synchronization
- Manage physical connection
- Bit rate control
- Line configuration
- Physical topology
- Transmission mode
- Multiplexing
- Switching

Layer 2: the data link layer

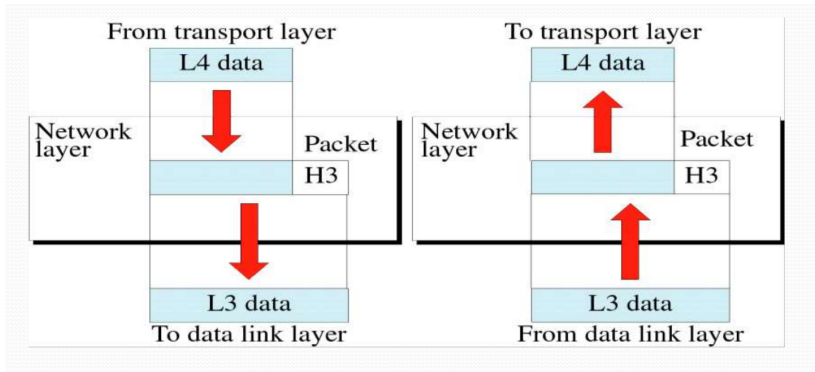
The data link layer is responsible for moving frames (=blocks of data) from one node to the next.



The functions of the data link layer are:

- Framing: dividing the data from the network layer into frames (blocks)
- Physical addressing: add a header to the frame to define the physical address (MAC) of the source and the destination machines
- Flow control: it is the traffic regulatory mechanism implemented by this layer that prevents the faster sender from drowning a slower receiver
- Error control: it provides the mechanism of error control in which it detects and re-transmits damaged or lost frames.
- Feedback: transmitting the frames, the system waits for the feedback.

The layer 3: the network layer

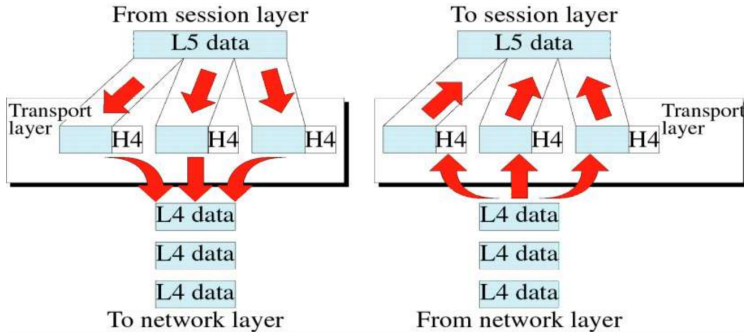


The functions of the network layer:

- It is responsible for the source to destination delivery of a packet across multiple networks.
- Routing: provide mechanism to transmit data over independent networks that are linked together (typically passing from router to router. Ex: from a local network to another local network passing through WLAN).
- Logical addressing: adds logical addresses of sender and receiver (require a new addressing system to permit transmission of data through independent networks).

The layer 4: the transport layer

The transport layer is responsible for source process to destination process delivery of entire message.



The transport layer provides two types of services:

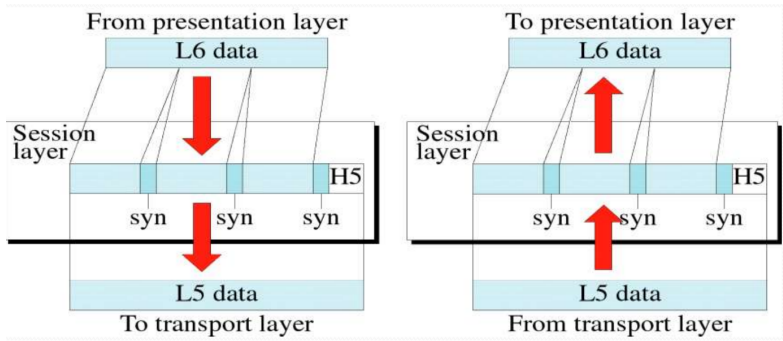
- ➊ Connection Oriented Transmission: In this type of transmission the receiving device sends an acknowledgment back to the source after a packet or group of packet is received.
- ➋ Connection-less Transmission: In this type of transmission the receiver does not acknowledge receipt of a packet.

The functions of the transport layer:

- Segmentation and Reassembly: Divide the message received from Session layer into Segments and number them to make a sequence for reassembly at the receiving side.
- Serviced point addressing: This layer makes sure that the messages is delivered to the correct process on destination machine.
- Error control: Make sure that the entire message arrives without errors else re-transmit.
- Flow control: Transport layer makes sure that the sender and the receiver communicate at a rate they both can handle.

The layer 5: the session layer

It is responsible for beginning, maintaining and ending the communication between two devices, which is called session.

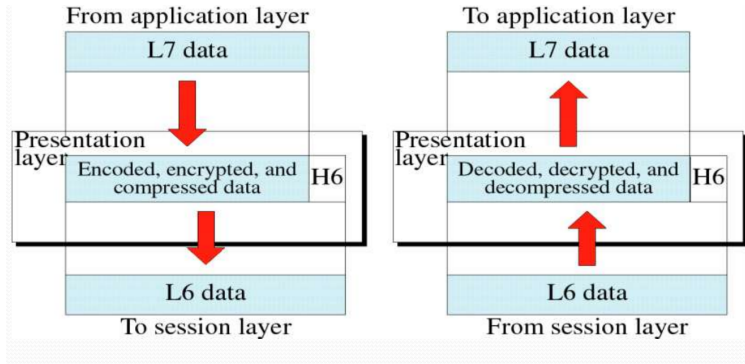


Functions of the session layer:

- Establishment, maintaining and ending a session:
 - ▶ Sends SYN packet - establish request
 - ▶ Receives ACK & SYN - established
 - ▶ To end - sender sends ACK
- Dialog Control: The session layer allows systems to enter into a dialog
- Synchronization: Allows a process to add checkpoints to a stream of data.

The layer 6: the presentation layer

This layer is concerned with the syntax and semantics of the information exchanged between two systems.



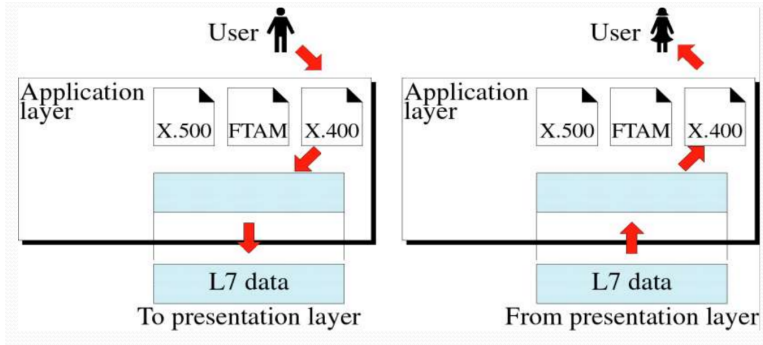
Functions of presentation layer:

- Data Translation: Encoding and Decoding Sender to Common format on Sending side Common to Receiving format on Receiver side
- Data Encryption: For security and privacy purpose
- Data Compression: Data compression reduces the number of bits contained in the information

In practice, few existing network models have a presentation layer ...

The layer 7: the application layer

It provides user interfaces and/or support for services, like e-mail, file transfer.



Functions of the application layer:

- Network Virtual terminal: It allows a user to log on to a remote host
- File Transfer Access, and Management: This application allows a user to access files in a remote host.
- Mail Services: This application provides various e-mail services.
- Directory Services: This application provides the distributed database sources and access for global information about various objects and services.

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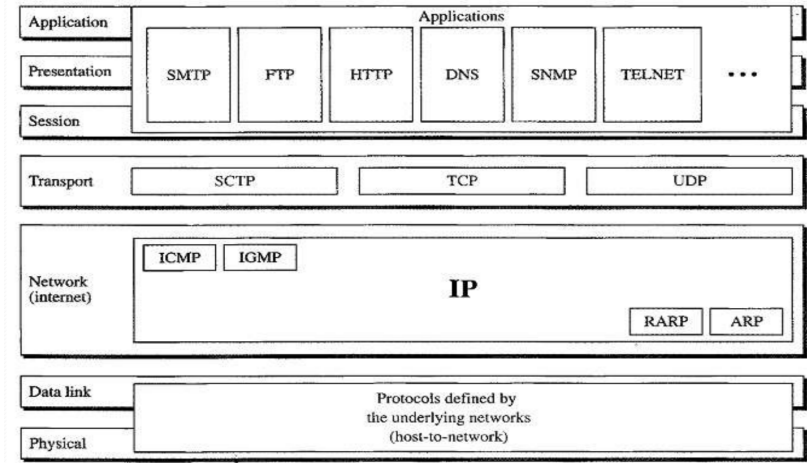
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The TCP/IP model ...

- ... forms the base of present day internet!
- ... is made of two protocols: TCP and IP
- ... is was initially used by ARPANET
- ... is made of four layers:
 - 1 Host-to-network
 - 2 Internet
 - 3 Transport
 - 4 Application

TCP/IP and OSI Model compared:



Layer 1: the host to network layer:

- It is the bottom layer of TCP/IP model also known as Network interface layer.
- The purpose of this layer is to connect the host and the network.

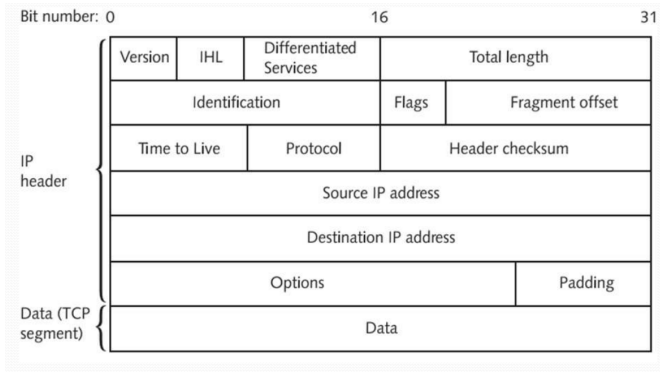
Layer 2: the internet layer

- Internet layer is similar to network layer of OSI model in functionality.
- This layer is responsible for delivering IP packets to their destinations.
- An important protocol of this layer is IP (Internet Protocol)

The internet protocol (IP)

- It is an unreliable and connection-less protocol
- IP transports data in packets called datagrams.
- IP does not keep track of the routes.

IP Datagram:



The layer 3: the transport layer

- Transport layer is similar to transport layer of OSI model
- Transport layer of TCP/IP model also provides connection oriented and connection-less services.
 - ▶ Connection Oriented - TCP (Transmission Control Protocol)
 - ▶ Connection-less UDP (User Datagram Protocol)

TCP:

- Transport layer used TCP for reliable connection oriented service
- The various function of TCP are:
 - ▶ Error control
 - ▶ Flow control
 - ▶ Sequencing

UDP:

- Transport layer uses this protocol for unreliable connection-less service.
- No assurance that all packets reached destination.
- No sequencing & No error checking.
- Useful in real time data transfer and quick transfer of large data (ex: video streaming, etc.)
- It follows the following principle: delivery is more important than accurate delivery!

Application layer:

- This layer is the combination of Application, Presentation and Session layer of the OS I model.
- This layer provides various services to different user applications.

More about the application layer:

- Application layer includes several high-level protocols that are used for wide variety of application like:
 - ▶ TELNET (Terminal Network): used for remote login.
 - ▶ SSH (Secured Shell): used for secured remote login.
 - ▶ FTP (File Transfer Protocol): For transfer of file from one system to another.
 - ▶ HTTP (Hyper Text Transfer Protocol): For fetching web pages on world wide web.
 - ▶ League Of Legend: For playing with people worldwide.
 - ▶ ...

Similarities between OSI & TCP:IP models:

- Both are based on the concept of a stack of independent protocols.
- Functionality of layers are roughly similar.
- Up to transport: network oriented.
- Above: user oriented.

Differences between OSI & TCP/IP models:

- OSI model has seven layers
 - ▶ TCP/IP has four layers
- OSI model provides clear distinction between services, interfaces and protocols.
 - ▶ TCP/IP doesn't provide clear distinction between services, interface and protocols.
- In OSI model transport layer is connection oriented
 - ▶ In TCP/IP transport layer is both connection oriented and connection-less (you can select)

More differences:

- In OSI the Data Link layer and the Physical layer are separate layers.
 - ▶ In TCP/IP the Data Link layer and Physical layer are combined into one: host-to-network layer.
- Protocols do not fit well into the OSI model
 - ▶ Protocols fit well with TCP/IP model
- Minimum size of OSI header is 5 bytes
 - ▶ In TCP/IP minimum size of the header is 20 bytes.